

02.1	Distance = $2\pi r$ OR πd seen OR Speed = $\frac{\text{distance}}{\text{time}}$ ✓ Speed = $11 \text{ (m s}^{-1}\text{)}$ ✓ cao	2	Award both marks for the correct answer
02.2	Velocity = $0 \text{ (m s}^{-1}\text{)}$ ✓	1	
02.3	Displacement = $2r = 84 \text{ m}$ ✓ Velocity = $7.0 \text{ (m s}^{-1}\text{)}$ ✓	2	Award both marks for the correct answer Condone 1 sf in final answer
Total		5	

Question	Marking guidance	Mark	Comments
	-1.6×10^{-19} (C) ✓ 1.66×10^{-27} (kg) ✓	2	Numbers all correct – 1 mark Accept 1.66×10^{-27} (kg)

24	D
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03	Positron correct 1.6×10^{-19} C electron Neutron correct zero or 0 antineutron Any 2 correct ✓ All 4 correct ✓	2	Accept +e Accept $\bar{n}$

04	ANY 2 from: Two photons ✓ Gamma (photons) ✓ Two photons must travel in opposite directions.✓ PLUS Gamma energy appropriate to the energy released by the annihilation of that amount of mass ✓ Opposite direction of the movement of the photons conserves momentum ✓	4	Accept two gamma rays must travel in opposite directions Accept whether or not kinetic energy of particles prior to collision is accounted for
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07.1	(electron) antineutrino, or $\bar{\nu}$ or $\bar{\nu}_e$ ✓	1	Accept mis-spellings when intention is clear but not confusion of neutrino and neutron. Accept anti electron neutrino
07.2	MAX 2 Only weak interactions with matter or very weakly ionising ✓ It has no charge ✓ It has a <u>very</u> small (or zero or negligible) mass ✓	2	
07.3	Actinium has one more proton or Z is 1 higher for actinium ✓ Actinium has one fewer neutron or A is the same for both ✓	2	Accept that the masses are the same

07.4	88 x 1.6x10⁻¹⁹ seen ✓ Expression for total mass or mass of protons seen ✓ Number of neutrons = 136 cao ✓	3	Expect to see $A \times 1.661 \times 10^{-27}$ or $88 \times 1.673 \times 10^{-27}$ but accept any values of nucleon mass (1.661×10^{-27}, 1.673×10^{-27} or 1.675×10^{-27}) Correct final answer gets all 3 marks
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Question	Marking guidance	Mark	Comments
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19	A	$^{22}_{10}\text{Ne}$ $^{23}_{11}\text{Na}$
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20	D	$e^{-} + e^{+} \rightarrow \gamma + \gamma$
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Question	Answers	Additional comments/Guidelines	Mark	AO
02	use of $\frac{\text{charge}}{\text{mass}}$ with a correct numerical substitution for either mass or charge ✓ $4.8 \times 10^7 \text{ ✓ (C kg}^{-1}\text{)}$	$Q = 2 \times 1.60 \times 10^{-19} \text{ (C)}$ $M = 4 \times 1.67 \times 10^{-27} \text{ or } 4 \times 1.66 \times 10^{-27}$ $\text{or } 4 \times 1.7 \times 10^{-27} \text{ (kg)}$ use of 1.7×10^{-27} gives $4.7 \times 10^7 \text{ C kg}^{-1}$	2	AO1 × 2
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	AO
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20	D	The mass is concentrated in the centre of the atom.	AO1
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21	B	zero 1.7×10^{-27}	AO1
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22	C	Momentum	AO1
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23	C	positron and neutrino	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	Doubling of energy AND use of 1.6×10^{-19} (J) seen with answer 1.63×10^{-13} (J) (≥ 3 sf) ✓		1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	Idea that excess energy is shared between the electron and the positron and the recoiling nucleus as kinetic energy ✓ Maximum KE that is available to each of the electron and the positron is 0.23×10^{-13} (J) to 0.25×10^{-13} (J) ✓	Condone lack of reference to recoiling nucleus Accept answer of 0.47×10^{-13} (J) shared between electron and positron	2	AO1 $\times$ 1 AO2 $\times$ 1
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
01	Volume of nucleus << volume of atom ✓ The nucleus has a charge ✓	Allow radius for volume Condone size Do not accept answers which refer to protons, neutrons or nucleons Do not accept the nucleus is very small Do not accept the nucleus is very dense	2	AO1
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	AO
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23	C	electron and positron	AO1
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17	C		206	82		AO3
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18	C	positron	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
03	Any two particles correct ✓ All three particles correct ✓✓ ${}^1_0n \rightarrow {}^1_1p + {}^{-1}_0e^{-} + {}^0_{0}\bar{\nu}$	Allow beta minus for e^{-} Condone missing minus sign on e Allow any order of particles	2	AO1
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO																
05	The mark scheme gives some guidance as to what statements are expected to be seen in a 1- or 2-mark (L1), 3- or 4-mark (L2) and 5- or 6-mark (L3) answer. Guidance provided in section 3.10 of the ‘Mark Scheme Instructions’ document should be used to assist in marking this question.		6	4 × AO1 2 × AO4																
	<table><tr><th>Mark</th><th>Criteria</th></tr><tr><td>6</td><td>All three areas covered in some detail. 6 marks can be awarded even if there is an error and/or parts of one aspect missing.</td></tr><tr><td>5</td><td>All three areas covered, at least two in detail. Whilst there will be gaps, there should only be an occasional error.</td></tr><tr><td>4</td><td>Two areas successfully discussed, or one discussed and two others covered partially. Whilst there will be several gaps, there should only be an occasional error.</td></tr><tr><td>3</td><td>One area discussed and one discussed partially, or all three covered partially. There are likely to be several errors and omissions in the discussion.</td></tr><tr><td>2</td><td>Only one area discussed or makes a partial attempt at two areas.</td></tr><tr><td>1</td><td>None of the three areas covered without significant error.</td></tr><tr><td>0</td><td>No relevant analysis.</td></tr></table>				Mark	Criteria	6	All three areas covered in some detail. 6 marks can be awarded even if there is an error and/or parts of one aspect missing.	5	All three areas covered, at least two in detail. Whilst there will be gaps, there should only be an occasional error.	4	Two areas successfully discussed, or one discussed and two others covered partially. Whilst there will be several gaps, there should only be an occasional error.	3	One area discussed and one discussed partially, or all three covered partially. There are likely to be several errors and omissions in the discussion.	2	Only one area discussed or makes a partial attempt at two areas.	1	None of the three areas covered without significant error.	0	No relevant analysis.
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Apparatus (1BP = partial, 3BP = full) - thin <u>gold</u> / <u>gold</u> foil with alpha particles directed towards it. - in an evacuated chamber / vacuum - microscope (to detect the alpha particles after deflection) OR fluorescent screen <i>This area is not addressed if there is no relevant diagram of apparatus</i>																				
Observations (2BP = partial, first 3BP full) - most alpha particles were not deflected - some were deflected by an angle of less than 90 degrees / small angles - a very few OWTTE were deflected by angles greater than 90 degrees / large angles - flashes / light observed where alpha particles hit the screen																				
What Rutherford deduced (1BP = partial, 3BP = full) - most are undeflected so atom is mostly empty space - some deflected by large angles so mass is located in the centre / very dense centre of atom - some deflected so nucleus is (positively) charged <i>Condone a deduction without explanation for partial only. Minimum 2 explanations required for full</i> Treat references to electrons as neutral																				

Total			0	
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15	C	3	AO1
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